Multigrid Finite State Projection for the Reaction–Diffusion Master Equation

Aditya Dendukuri

University of California, Santa Barbara

The problem: rare events in spatial stochastic kinetics

Reaction–diffusion on a lattice: molecules react within a voxel and hop to neighbours. At low copy number the noise is intrinsic.

- Bistable chemistry gives **metastable** basins; the relevant transitions (basin switching, nucleation) are **rare**.
- Target: the stationary distribution *and* the switching **rate**, not a single trajectory.

Limitations of existing tools:

approach	limitation
SSA (sampling)	rare switch requires prohibitively many samples; λ_2 underflows
Dense / direct FSP	joint is $n_{\max}^{K^2}$ ($200\times200 \Rightarrow n_{\max}^{40000}$); LU infeasible by 6.8M states
Standard multi-grid (uniform P)	coarsens, but corrupts the slow mode: 150% rate error

This talk

One operator $A_c = RAP$ on **two axes**: count (basins) and space (voxel blocks). Solve the joint at scale, *and* reduce it to a $\sim$200-state rate-faithful model.

The Chemical Master Equation

Distribution $p(n,t)$ over integer counts, one linear ODE:

$$\dot{p} = A\,p.$$

- $\mathbf{1}^\top A = 0$ (mass-conserving); steady state = null vector.
- p is the full distribution: tails, multiple peaks, rare events.

Finite State Projection

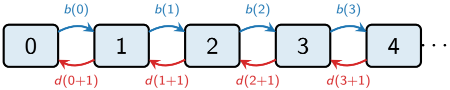
Restrict to a finite support $\mathcal{S}$ and integrate

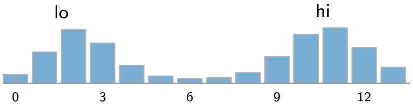
$$p(t+\Delta t) = e^{\Delta t\, A_{\mathcal{S}}}\, p \quad (\texttt{expv}).$$

Expand $\mathcal{S}$ where probability flows out, discard the tail. Cost $\sim |\mathcal{S}|$.

Bistable birth–death (Schlögl):

$2X \rightleftharpoons 3X, \quad \emptyset \rightleftharpoons X$ (state-dependent $b(n), d(n)$)





stationary $p(n)$: two peaks

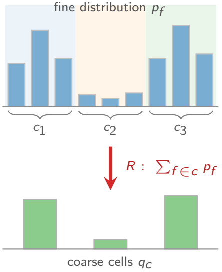
Restriction R: sum fine mass into cells

Definition 1 (Partition & restriction)

Partition the fine support into disjoint cells $\mathcal{S} = \bigsqcup_c \mathcal{S}_c$; each microstate $f \in \mathcal{S}$ lies in exactly one cell c. The restriction $R \in \mathbb{R}^{|\mathcal{C}| \times |\mathcal{S}|}$ is

$$R_{cf} = \mathbf{1}[f \in c], \qquad (R\,p)_c = \sum_{f \in c} p_f.$$

The coarse variable is the cell label. A cell is *any* group of microstates: a count-basin, a block of voxels, an aggregate. The example below uses three arbitrary cells c_1, c_2, c_3.



$(R\,p)_c = \sum_{f \in c} p_f$ (cell-wise sum)

Prolongation P: distribute by the within-cell conditional

Definition 2 (Prolongation)

Fix within-cell weights $\varphi_f = \pi(f \mid c)$, the *stationary conditional* ($\sum_{f\in c} \varphi_f = 1$). Then $P \in \mathbb{R}^{|\mathcal{S}|\times|\mathcal{C}|}$ is

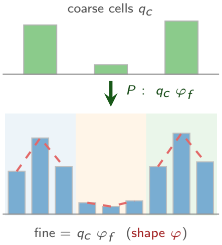
$$P_{fc} = \varphi_f \, \mathbf{1}[f \in c], \qquad (P\,q)_f = q_c\,\varphi_f \quad (f \in c).$$

Galerkin coarse generator $A_c := R\,A\,P$.

Lemma 1 (Consistency)

$R\,P = I_{|\mathcal{C}|}$ and $\mathbf{1}^\top A_c = 0$. ($\|RP - I\| = 2.2\mathrm{e}{-16}$.)

Taking $\varphi = \pi(\cdot \mid c)$ is the **structure-preserving** (stationary-weighted) choice: it restores each cell's conditional shape, so multimodal structure survives $P \circ R$. Uniform weights $\varphi_f = 1/|\mathcal{S}_c|$ do *not* preserve the slow dynamics (later: the rate).



$(P\,q)_f = q_c\,\varphi_f, \;\; \varphi_f = \pi(f \mid c)$

Operator splitting and the V-cycle

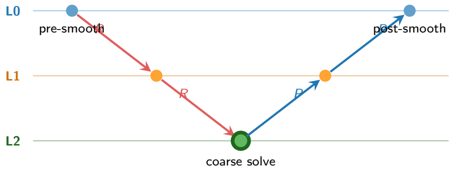
$$A = A_{\text{fast}} + A_{\text{slow}}$$

- A_{fast}: within-basin relaxation, large $|\lambda| \to$ **smoother** S
- A_{slow}: basin switching $\to$ **coarse grid** A_c

S damps the fast error from A_{fast}; the coarse solve on A_c corrects the slow error left behind. The V-cycle alternates the two, for the backward-Euler step $(I - \Delta t\, A)x = b$:

1. pre-smooth $x \leftarrow S^{\nu} x$
2. residual $r = b - (I - \Delta t\, A)x$
3. coarse solve $(I - \Delta t\, A_c)\, e_c = R\, r$
4. correct $x \leftarrow x + P\, e_c$
5. post-smooth $x \leftarrow S^{\nu} x$

Recurse on A_c for a full V-cycle.



Measured (bistable Schlögl, K=5)

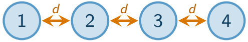
coarsening $8547 \to 6$ ($1424\times$); contraction 0.75/cycle; residual $3.5\mathrm{e}{-2} \to 3.1\mathrm{e}{-5}$ in 25 cycles; rel. err vs direct 1.5×10^{-4}.

The spatial problem: the reaction–diffusion master equation

$K \times K$ grid; voxel $\mathbf{k}$ holds count $n_{\mathbf{k}}$. Diffusion is a **first-order (linear) reaction $X_{\mathbf{k}} \to X_{\mathbf{l}}$** at rate $d = D/h^2$, so space coarsening is *exact* (Theorem 1).

$$A = \underbrace{\textstyle\sum_{\mathbf{k}} A^{\mathrm{rx}}_{\mathbf{k}}}_{\text{reaction (diag.)}} + \underbrace{\textstyle\sum_{\langle \mathbf{k},\mathbf{l}\rangle} A^{\mathrm{diff}}_{\mathbf{kl}}}_{X_{\mathbf{k}} \to X_{\mathbf{l}}\ \text{(off-diag.)}}$$

- **Joint** law, up to $n_{\max}^{K^2}$; a 200×200 grid is $n_{\max}^{40000}$, never assembled.
- Neighbour-only coupling ⇒ smooth, near-steady away from the front.



1D voxel chain (K=4): nodes react, edges diffuse

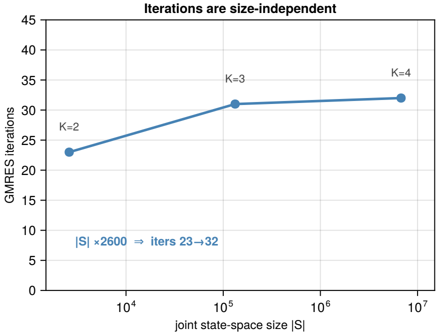
$$\begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{bmatrix} A^{\mathrm{rx}} & d & \multicolumn{2}{c}{0} \\ d & A^{\mathrm{rx}} & d & \\ & d & A^{\mathrm{rx}} & d \\ \multicolumn{2}{c}{0} & d & A^{\mathrm{rx}} \end{bmatrix} \end{array}$$

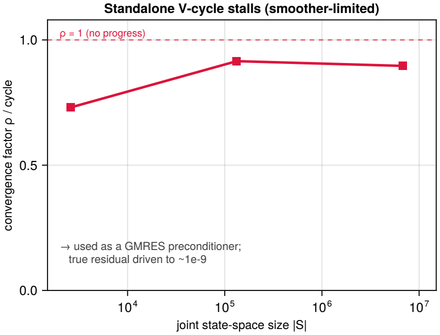
diagonal = per-voxel CME (A^{rx}); off-diagonal = diffusion (d)

Role of the coarse operator

One operator $A_c = RAP$ serves two purposes: it solves the joint at scale, and reduces the model to $\sim$200 states that preserve the switching rate.

Bounded iteration count across state-space size





K	$\lvert\mathcal{S}\rvert$	levels	GMRES its
2	2,601	2	23
3	132,651	5	31
4	**6,765,201**	8	**32**

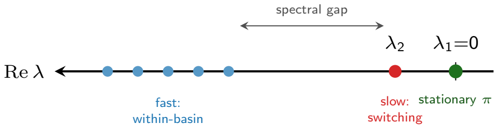
Ring joint (RDME): bistable Schlögl, K-voxel ring. $\lvert\mathcal{S}\rvert \times 2600$, iterations $23 \to 32$, size-independent. 6.8M states solved to $6\mathrm{e}{-9}$ in 55 s; direct LU infeasible.

Construction: double-pairwise aggregation + plain Galerkin (preserves the M-matrix) + V-cycle-preconditioned GMRES.

The metastable rate is an eigenvalue

Spectrum of A: $0 = \lambda_1 > \mathrm{Re}\,\lambda_2 \geq \cdots$; the stationary law π is the $\lambda_1{=}0$ mode.

- Bistability gives a **spectral gap**: one eigenvalue λ_2 lies near 0, isolated from the bulk.
- Its mode is the basin imbalance; relaxation $p(t) - \pi \sim e^{\lambda_2 t}$.
- $|\lambda_2|$ *is* the basin-switching rate, $\approx 1/\langle T_{\mathrm{switch}}\rangle$: the rare event SSA cannot sample.

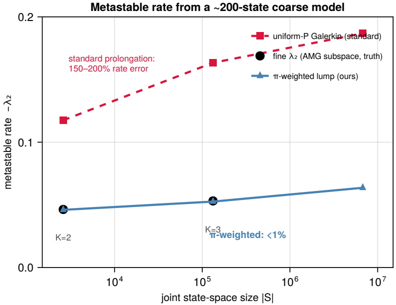


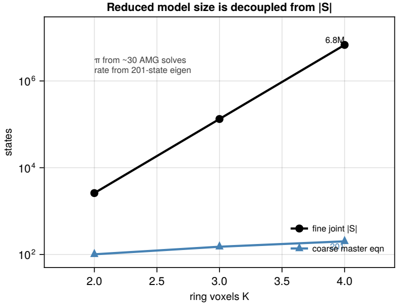
isolated $\lambda_2 \Rightarrow$ metastability; the bulk is the fast within-basin relaxation

Experiment

For each ring size K, compute λ_2 on (i) the fine joint A (truth, feasible only at small K) and (ii) the $\sim$200-state coarse operator $A_c = RAP$, with π-weighted vs uniform P.

Recovering the rare-event rate from ∼200 states





K	$\lvert\mathcal{S}\rvert$	coarse	$-\lambda_2$	rel. err
2	2,601	101	0.0460	0.65%
3	132,651	151	0.0526	0.99%
4	**6,765,201**	201	**0.0636**	n/a

K=4: fine ref. infeasible; rate *obtained* from 201 states. (uniform-P yields 0.187.)

Reduce the ring onto total count $Y = \sum_v n_v$: a ∼200-state coarse master equation reproduces λ_2 to $< 1\%$, but only with **stationary-weighted** $\pi(s \mid Y)$; uniform-P errs by 150%.

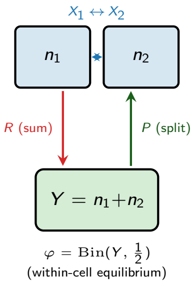
π is the λ=0 (stationary) mode (∼30 AMG solves): **no rare transition sampled** (unlike SSA).

Coarsening the space axis: merge voxels

Counts coarsened; now coarsen *space*. Diffusion (the linear hop $X_i \to X_j$) equilibrates quickly within a region; the slow mode is the coarse profile.

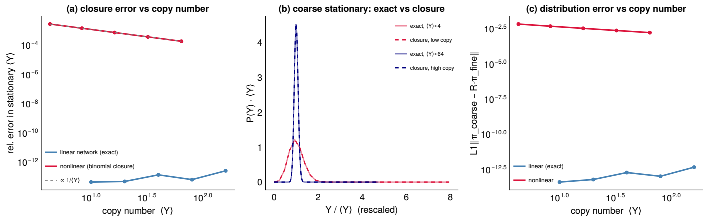
Same $A_c = R\,A\,P$, cell = **block of voxels**:

- coarse variable $Y = \sum_{v \in c} n_v$.
- per-voxel weights $\varphi_f = \pi(\cdot \mid c)$, as before.
- affine network: φ is **binomial / multinomial** in closed form (no solve).
- nonlinear (Schlögl ring): φ is the AMG stationary conditional.



Theorem 1 (Exact coarsening for linear networks)

If every reaction propensity is affine in the counts and diffusion is linear, then the within-cell conditional $\varphi = \pi(\cdot \mid c)$ is exactly (multi)nomial and the Galerkin operator $A_c = R\,A\,P$ generates the coarse variable *exactly*: the closure error is zero.



Nonlinear α. For a smooth propensity the leading closure error is

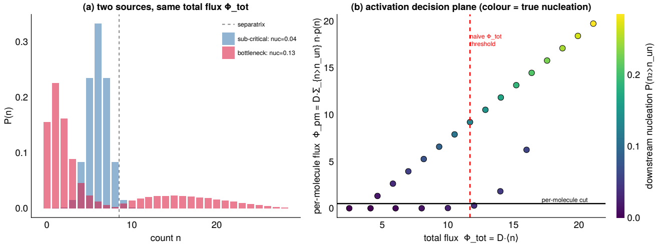
$$\tfrac{1}{2}\,\alpha''(\langle n\rangle)\,\mathrm{Var}(n) = O\big(1/\langle n\rangle\big) \xrightarrow[\text{high copy}]{} 0.$$

- **Linear:** exact to machine precision, every copy number.
- **Nonlinear:** error $\propto 1/\langle n\rangle$.
- Coarsen **where counts are high** (bulk); keep the front fine.

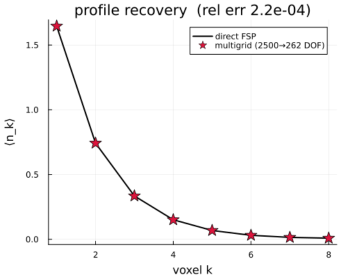
Space axis: coarsen the bulk, keep the fine grid where the front moves.

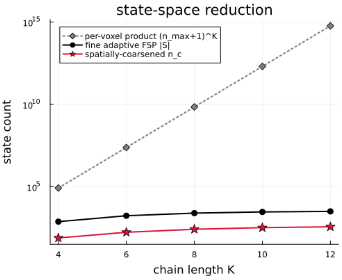
$$\Phi_{\mathbf{k}} = D\,\langle n\rangle \quad \text{(total flux)}, \qquad \phi_{\mathbf{k}} = D \sum_{n>n_{\mathrm{un}}} n\,p(n) \quad \text{(per-molecule flux)}$$

- Φ misses a **bottleneck**: low mean, transition-competent tail.
- ϕ counts only molecules above n_{un}, those that nucleate a neighbour.
- **refine if Φ or $\phi > \mathrm{tol}$.**



1D birth–death–diffusion: exact two-grid

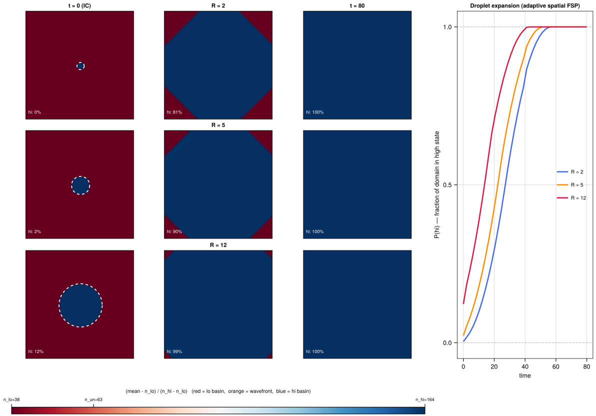




Linear chain $\Rightarrow$ coarsening is **exact**, with no closure error.

- coarse solve matches the fine FSP: rel. err 2.2×10^{-4}
- full joint $\sim 10^{15}$ at $K=12$; FSP support plateaus ~ 3000
- $|\mathcal{S}|$ tracks the active band, not K

Bistable Schlögl: critical droplet on a 60×60 grid



Schlögl RDME, 60×60, $D=2$; initial radii $R=2,5,12$. Colour = $P(n > n_{\mathrm{un}})$ (red lo, white front, blue hi). **Right:** high-basin fraction vs. time, all radii grow. Bulk voxels frozen at a basin; only the thin front ring evolves.

Summary

- **One principle, two axes.** Merging voxels and grouping count-basins are the *same* coarsening; multigrid applies either, recursively.
- **Known prolongation.** $P =$ the within-cell steady state; $A_c = RAP$ is **exact for linear networks**, with $O(1/\langle n \rangle)$ error at high copy number.
- **Adaptive grid.** Total *and* per-molecule flux keep the front fine and the bulk coarse; $|\mathcal{S}|$ tracks the active band, not the domain.
- **Measured & scalable.** A complete V-cycle: $1424\times$ coarsening, 1.5×10^{-4} vs direct; exact on the linear chain. As a GMRES preconditioner it is **size-independent**: $23 \to 32$ iterations as $|\mathcal{S}|$ grows $2.6\mathrm{k} \to 6.8\mathrm{M}$.
- **Scope.** Exact under timescale separation; degrades when correlations extend (a broad 2D interface), where the slow subspace is no longer low-dimensional.